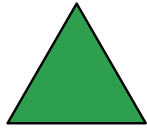
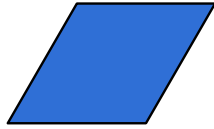


**Problem 1:** Your partner picks one of these four blocks in secret. Can you always find out which one with two yes-or-no questions?



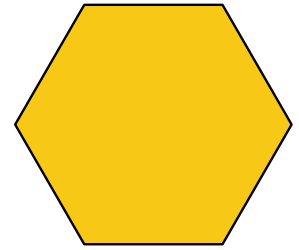
green triangle



blue rhombus

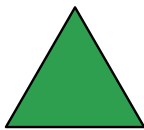


red trapezoid

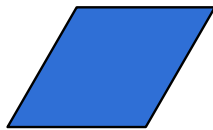


yellow hexagon

**Problem 2:** Your partner picks one of these eight blocks in secret. You can only point and ask “Is it this one?” What is the most questions you could need before you are sure?



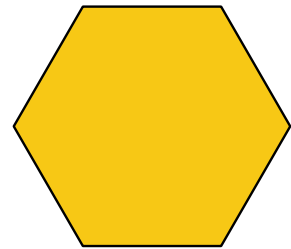
green triangle



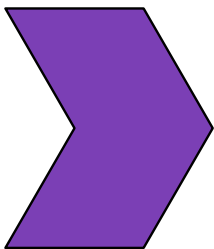
blue rhombus



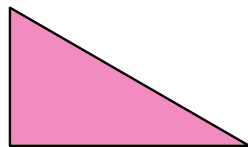
red trapezoid



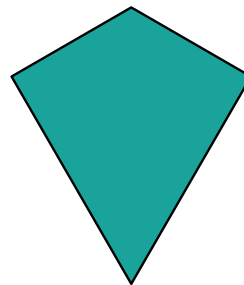
yellow hexagon



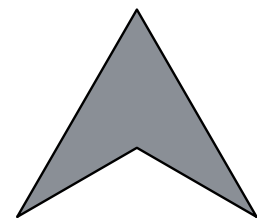
purple chevron



pink right triangle

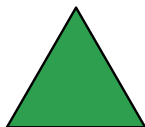


teal kite



gray dart

**Problem 3:** Your partner picks one of these eight blocks in secret. Can you always find out which one with three yes-or-no questions?



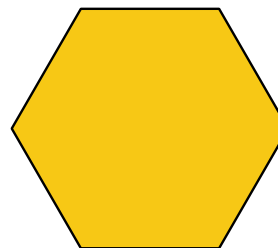
green triangle



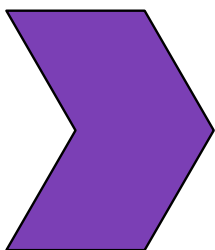
blue rhombus



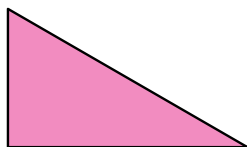
red trapezoid



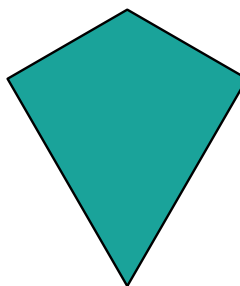
yellow hexagon



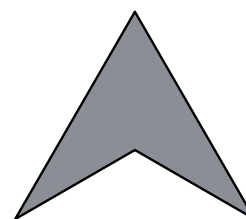
purple chevron



pink right triangle



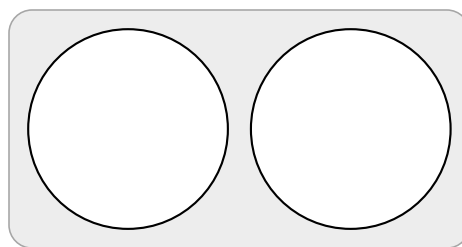
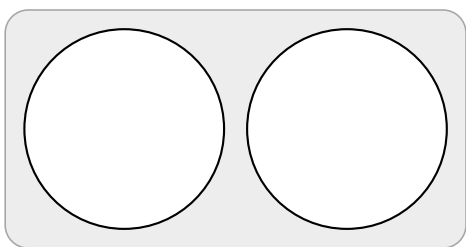
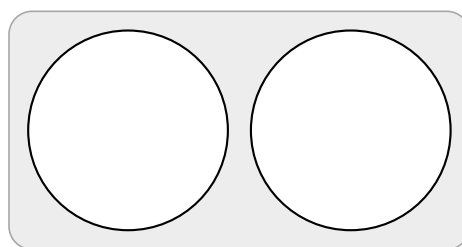
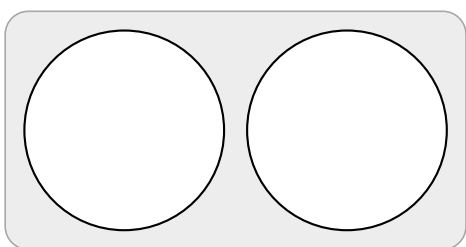
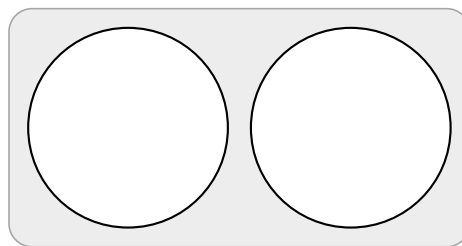
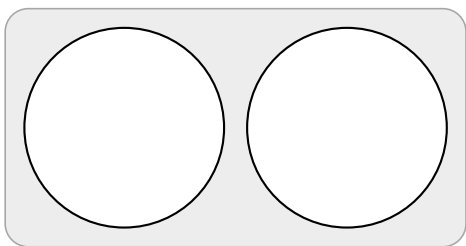
teal kite



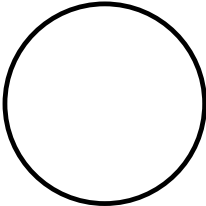
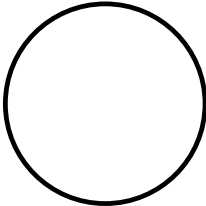
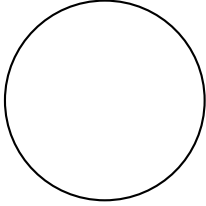
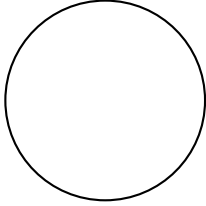
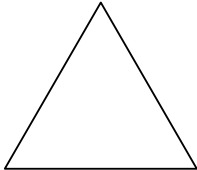
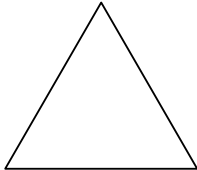
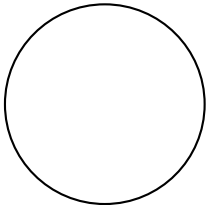
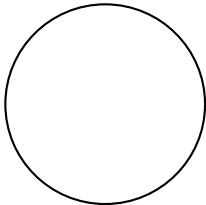
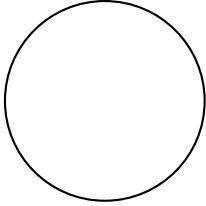
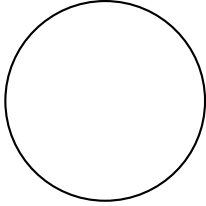
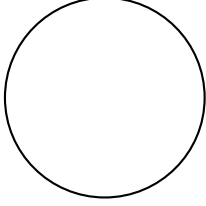
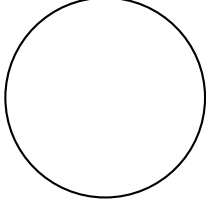
gray dart

**Problem 4:** Now you pick one of the eight blocks in secret, and your partner may ask only two yes-or-no questions. Can your partner always be sure which block you picked?

**Problem 5:** A secret is two counters in a row, each one red or yellow. How many different secrets can you make?



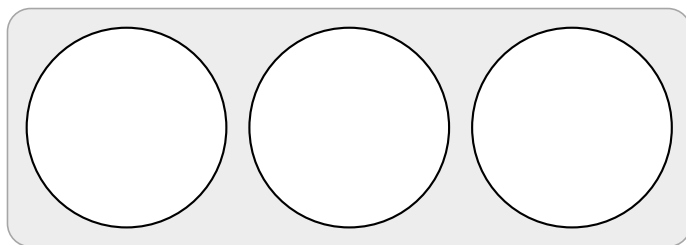
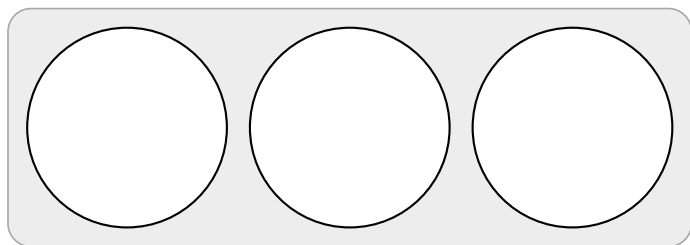
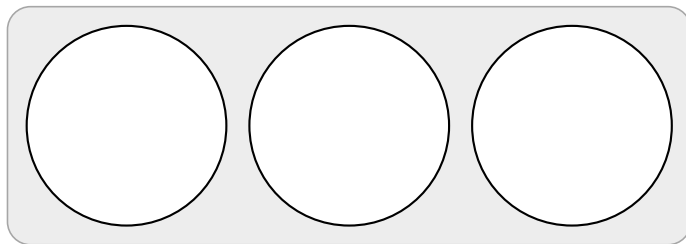
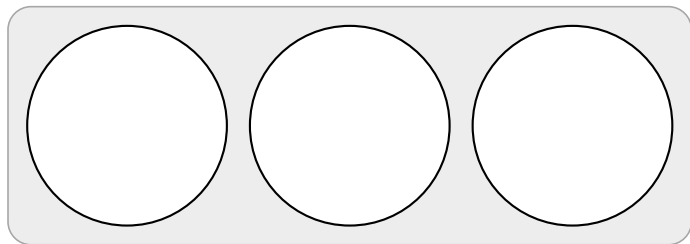
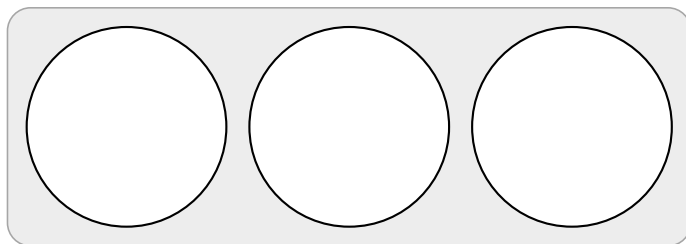
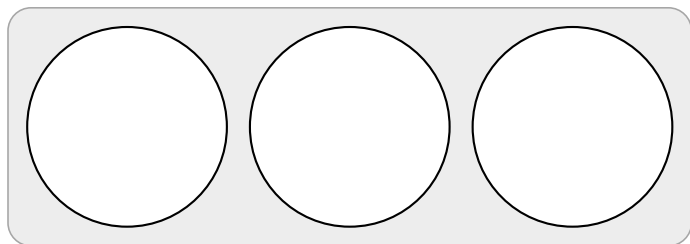
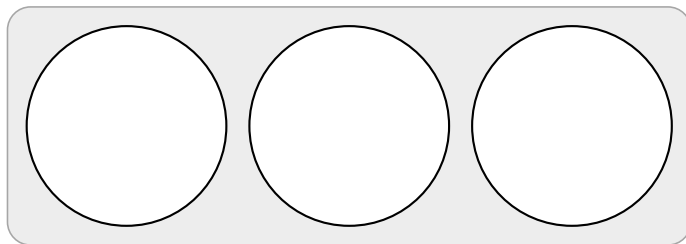
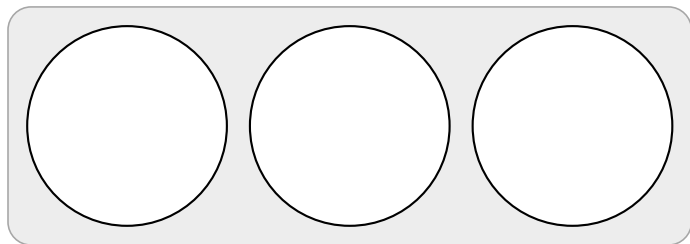
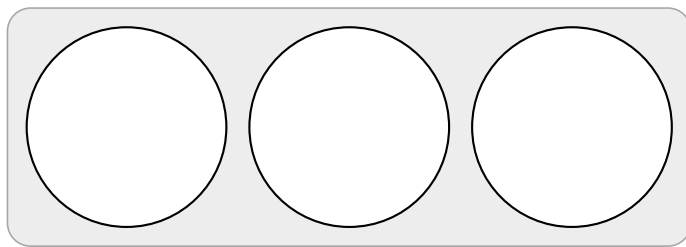
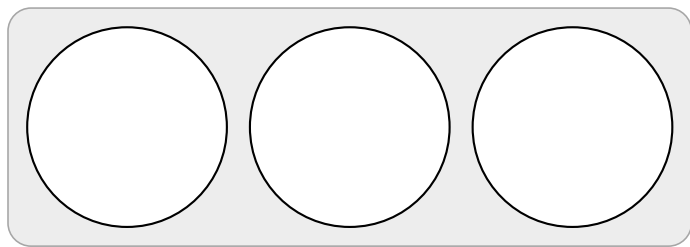
**Problem 6:** The grown-up hides two counters behind the folder. Put two counters in a test row to make a test. The grown-up puts a green triangle under each counter that matches the secret and tells you how many. When you say what the secret is, the grown-up lifts the folder and you check every score together. Can you always be sure of the secret after one test? Can you always be sure after two tests?

|                                             |                                                                                     |                                                                                      |                                                                                          |
|---------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| secret                                      |    |    |                                                                                          |
| copy of test                                |    |    |                                                                                          |
|                                             |   |    |                                                                                          |
| <hr style="border-top: 1px dashed black;"/> |                                                                                     |                                                                                      |                                                                                          |
|                                             |                                                                                     |                                                                                      | folder                                                                                   |
| test                                        |  |  | <div style="border: 1px solid black; width: 120px; height: 80px; margin: 0 auto;"></div> |
|                                             |  |  | <div style="border: 1px solid black; width: 120px; height: 80px; margin: 0 auto;"></div> |
|                                             |  |  | <div style="border: 1px solid black; width: 120px; height: 80px; margin: 0 auto;"></div> |

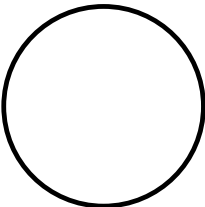
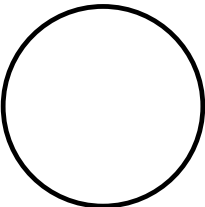
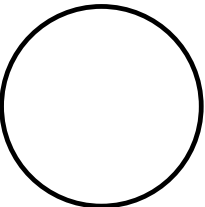
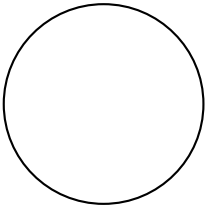
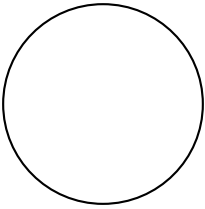
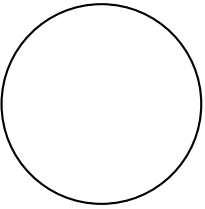
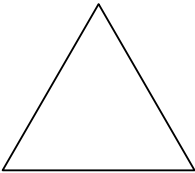
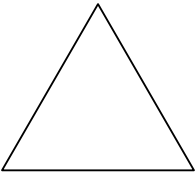
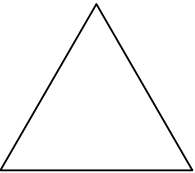
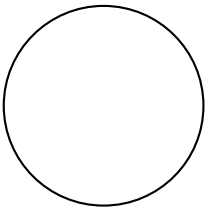
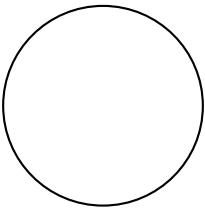
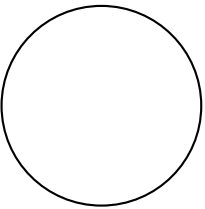
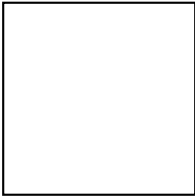
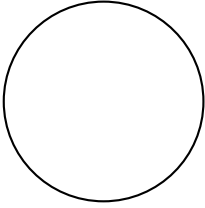
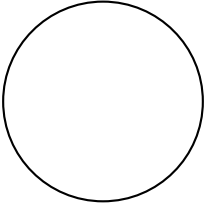
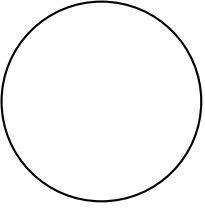
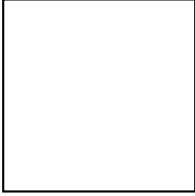
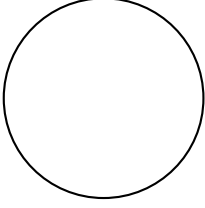
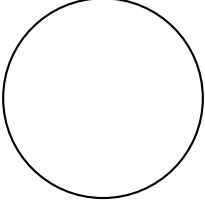
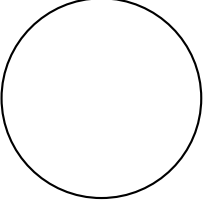
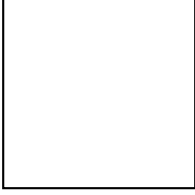
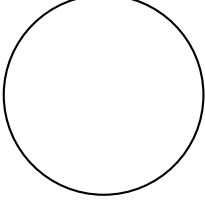
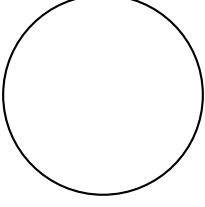
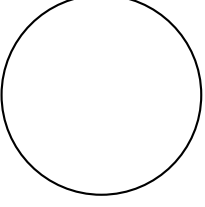
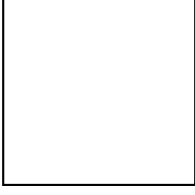
**Problem 7:** For each row of tests and scores, which secrets could the grown-up be hiding? Put counters on every one.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                            |
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| <hr style="border: 0; border-top: 1px solid gray; margin: 10px 0;"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                            |
| <div style="display: flex; align-items: center; gap: 10px;"> <div style="display: flex; gap: 5px;"> <div style="background-color: red; color: white; border-radius: 50%; width: 30px; height: 30px; display: flex; align-items: center; justify-content: center;">R</div> <div style="background-color: yellow; color: black; border-radius: 50%; width: 30px; height: 30px; display: flex; align-items: center; justify-content: center;">Y</div> </div> <div style="border: 1px solid black; padding: 5px; width: 40px; text-align: center;">0</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <div style="display: flex; gap: 20px;"> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> </div> | <div style="display: flex; gap: 20px;"> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> </div> |
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| <div style="display: flex; align-items: center; gap: 10px;"> <div style="display: flex; gap: 5px;"> <div style="background-color: red; color: white; border-radius: 50%; width: 30px; height: 30px; display: flex; align-items: center; justify-content: center;">R</div> <div style="background-color: red; color: white; border-radius: 50%; width: 30px; height: 30px; display: flex; align-items: center; justify-content: center;">R</div> </div> <div style="border: 1px solid black; padding: 5px; width: 40px; text-align: center;">1</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <div style="display: flex; gap: 20px;"> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> </div> | <div style="display: flex; gap: 20px;"> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> </div> |
| <hr style="border: 0; border-top: 1px solid gray; margin: 10px 0;"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                            |
| <div style="display: flex; flex-direction: column; align-items: center; gap: 10px;"> <div style="display: flex; gap: 5px;"> <div style="background-color: red; color: white; border-radius: 50%; width: 30px; height: 30px; display: flex; align-items: center; justify-content: center;">R</div> <div style="background-color: red; color: white; border-radius: 50%; width: 30px; height: 30px; display: flex; align-items: center; justify-content: center;">R</div> </div> <div style="border: 1px solid black; padding: 5px; width: 40px; text-align: center;">1</div> </div> <div style="display: flex; flex-direction: column; align-items: center; gap: 10px;"> <div style="display: flex; gap: 5px;"> <div style="background-color: yellow; color: black; border-radius: 50%; width: 30px; height: 30px; display: flex; align-items: center; justify-content: center;">Y</div> <div style="background-color: red; color: white; border-radius: 50%; width: 30px; height: 30px; display: flex; align-items: center; justify-content: center;">R</div> </div> <div style="border: 1px solid black; padding: 5px; width: 40px; text-align: center;">0</div> </div> | <div style="display: flex; gap: 20px;"> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> </div> | <div style="display: flex; gap: 20px;"> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> </div> |
| <hr style="border: 0; border-top: 1px solid gray; margin: 10px 0;"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                            |
| <div style="display: flex; flex-direction: column; align-items: center; gap: 10px;"> <div style="display: flex; gap: 5px;"> <div style="background-color: red; color: white; border-radius: 50%; width: 30px; height: 30px; display: flex; align-items: center; justify-content: center;">R</div> <div style="background-color: yellow; color: black; border-radius: 50%; width: 30px; height: 30px; display: flex; align-items: center; justify-content: center;">Y</div> </div> <div style="border: 1px solid black; padding: 5px; width: 40px; text-align: center;">1</div> </div> <div style="display: flex; flex-direction: column; align-items: center; gap: 10px;"> <div style="display: flex; gap: 5px;"> <div style="background-color: red; color: white; border-radius: 50%; width: 30px; height: 30px; display: flex; align-items: center; justify-content: center;">R</div> <div style="background-color: red; color: white; border-radius: 50%; width: 30px; height: 30px; display: flex; align-items: center; justify-content: center;">R</div> </div> <div style="border: 1px solid black; padding: 5px; width: 40px; text-align: center;">0</div> </div> | <div style="display: flex; gap: 20px;"> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> </div> | <div style="display: flex; gap: 20px;"> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> <div style="border: 1px solid black; border-radius: 50%; width: 100px; height: 100px;"></div> </div> |

**Problem 8:** Now a secret is three counters in a row. How many different secrets can you make?



**Problem 9:** The grown-up hides three counters. Play the same game. Can you always be sure of the secret after three tests?

|                                             |                                                                                     |                                                                                     |                                                                                      |                                                                                       |
|---------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| secret                                      |    |    |    |                                                                                       |
| copy of test                                |    |    |    |                                                                                       |
|                                             |    |    |    |                                                                                       |
| <hr style="border-top: 1px dashed black;"/> |                                                                                     |                                                                                     |                                                                                      |                                                                                       |
|                                             |                                                                                     |                                                                                     |                                                                                      | score                                                                                 |
| test                                        |  |  |  |  |
|                                             |  |  |  |  |
|                                             |  |  |  |  |
|                                             |  |  |  |  |